The empirical recurrence for OEIS A250991: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A250991 records "Empirical recurrence of order 51", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 457$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 51, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 51 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A250991 is "Number of (n+1)X(5+1) 0..2 arrays with nondecreasing min(x(i,j),x(i,j-1)) in the i direction and nondecreasing absolute value of x(i,j)-x(i-1,j) in the j direction.". It has offset 1 and begins

$$8298, 67409, 456514, 2800667, 16381510, 92713991, 514669302, 2820364768, \ \ldots$$

The entry states:

> Empirical recurrence of order 51 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 12:57:15, revision 6) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 729$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 457$ of the 729, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 457.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1458$ exact terms of the model returns order 51 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{51} c_i\, a(n-i),$$

c_1	62	c_{14}	−167401325575	c_{27}	−592108725519418	c_{40}	−1106380932621500
c_2	−1769	c_{15}	−651326639628	c_{28}	−3265596970594458	c_{41}	1798503127323800
c_3	30554	c_{16}	2690124640228	c_{29}	2717341984866066	c_{42}	724102025510924
c_4	−352038	c_{17}	−1625798052792	c_{30}	4252003590751019	c_{43}	−529999156757896
c_5	2785446	c_{18}	−11872755734793	c_{31}	−5598184080797116	c_{44}	−332611556852576
c_6	−14633191	c_{19}	28192354428562	c_{32}	−4096128745930403	c_{45}	64455897704480
c_7	41403938	c_{20}	10648271103766	c_{33}	7801413897908610	c_{46}	89418459660400
c_8	39117426	c_{21}	−128857117750986	c_{34}	3074628766657150	c_{47}	12955442331008
c_9	−954110488	c_{22}	117508409389595	c_{35}	−8099327637325228	c_{48}	−8919511867200
c_{10}	4045088269	c_{23}	299938152191356	c_{36}	−2039203730120696	c_{49}	−4225336636800
c_{11}	−4761013118	c_{24}	−642472339984822	c_{37}	6475667012190628	c_{50}	−721609920000
c_{12}	−28045091326	c_{25}	−265434438384648	c_{38}	1449117347295544	c_{51}	−45921600000
c_{13}	140289017422	c_{26}	1783982535816027	c_{39}	−3978142947593814		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 51, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $51+4$ terms removed returns order 51 as well, so $\deg q_{\min} = 51 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $51+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 51 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A250991.

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